Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

ML engineer and fraud data analyst with **8+ years building detection systems on large-scale financial datasets**. Built **Isolation Forest anomaly detection** at GMU across 3 research pipelines - surfaced behavioral deviations that manual review missed. Applied **Python ML pipelines to 1.6M financial records** at Entrata with output validation and compliance gates; **100% PHI compliance** on AWS migration with zero incidents. Expert in Python, SQL, and R; advanced in scikit-learn, XGBoost, Random Forest, and Tableau. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Anomaly Detection (Isolation Forest) | Fraud Signal Detection | Python | SQL | Random Forest | XGBoost | scikit-learn | Feature Engineering | Large Dataset Analysis | Risk Segmentation | Tableau / Power BI | AWS / PHI Compliance

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Research Associate - ML Detection & Data Analytics

Fairfax, Virginia

- Built **Isolation Forest anomaly detection system** (Python) across 3 research pipelines - surfaced behavioral anomalies that manual review missed, **cut manual QA 70%**, improved downstream model reliability.
- Built regression models (**R^2 = 0.977**) across WHO and UK Road Safety datasets spanning **190+ countries** - multi-source data collection, transformation, validation, and stakeholder reporting.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - improved research comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer - Financial Data Pipelines & Compliance

Pune, India

- Processed **1.6M financial records** through Python and OpenAI API transformation pipelines - production ML on regulated data with output validation, quality gates, and compliance checkpoints.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines - **zero security incidents**; built 5+ internal tools cutting resolution time by 40%.
- Tuned **200+ SQL queries** across PostgreSQL and MySQL - **30% production performance gain** across all client environments; containerized 30%+ of microservices fleet with Docker (50% execution time reduction).
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation across 120+ Agile sprints - **100K+ MAU, 99.5% uptime** sustained over 6+ months.
- Conducted 300+ peer code reviews enforcing secure coding practices, TDD/BDD standards, and data quality standards across all major releases.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** building 20+ full-stack data modules (Python, PostgreSQL) - 99.5% uptime and 25% faster report generation.

PROJECTS

Global Health & Road Safety - Anomaly Detection & ML Pipelines
2024-2025 | Python, R, Isolation Forest, scikit-learn, Tableau, ggplot2, Quarto | George Mason University

- Built **AI-powered anomaly detection auditor (Isolation Forest)** flagging behavioral deviations across 3 active research projects - cut manual QA by **70%**; same detection methodology applied to transaction fraud and AML pattern analysis.
- Built end-to-end data preprocessing and regression pipelines (R^2 = 0.977) across **190+ countries (WHO + UK datasets)** - multi-source ingestion, transformation, validation, output verification, and KPI dashboards.

CrashClarity - Large-Scale Pattern Detection (104K Records)
2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering | Live: Portfolio

- Achieved **85%+ accuracy** on 104K+ records by building a Random Forest + Logistic Regression pipeline with feature engineering - multi-variable risk segmentation and interactive dashboard for non-technical stakeholders.

Real Estate Analysis - Risk Factor Modeling (XGBoost)
2025 | Python, XGBoost, Random Forest, Feature Engineering, Pandas

- Analyzed socioeconomic risk factors across 20+ variables (Connecticut property data 2001-2022) using **XGBoost and Random Forest** - identified key risk predictors and segmentation drivers.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, anomaly detection, ML systems, data pipelines, risk analysis

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

ML & Detection: Isolation Forest, Random Forest, XGBoost, Logistic Regression, scikit-learn, Feature Engineering, Statistical Modeling, Anomaly Detection, NumPy, Pandas

Data & Infra: PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Flask, RabbitMQ, Microservices **Visualization:** Tableau, Power BI, ggplot2, Plotly, Quarto

Compliance: PHI/HIPAA compliance, Libsodium encryption, audit logging, secure coding, TDD/BDD

AI Tools: LangChain, LangGraph, OpenAI API, BERT, spaCy, PyTorch, TensorFlow